

# Chris Dai

✉ yizhed7@gmail.com |  |  chriszyd.github.io

## Education

### Harvard University

M.S. Health Data Science, T.H. Chan School of Public Health

### University of Michigan, Ann Arbor

B.S.E. Industrial and Operations Engineering; Minor in Computer Science

### UM-SJTU Joint Institute, Shanghai Jiao Tong University

B.S.E. Electrical and Computer Engineering

Dean's Honor List (2024, 2025) · Myun Lee Scholarship (2025)

Relevant Coursework: Data Structures & Algorithms, Database Systems, Machine Learning, Optimization Methods, Linear Algebra

Aug. 2026 – Expected May 2028

Boston, Massachusetts, USA

Aug. 2024 – May 2026

Ann Arbor, Michigan, USA

Sep. 2022 – Aug. 2026

Shanghai, China

## Skills

**Engineering:** Python, TypeScript, JavaScript, FastAPI, Flask, React, Node.js, Next.js, Playwright / Chrome DevTools Protocol

**ML & Data:** PyTorch, K-means / PCA, time-series feature engineering, SQL, PostgreSQL, dbt, evaluation-harness design

**LLM & Evaluation:** SFT corpus construction, synthetic dialogue generation & data flywheels, LLM-as-judge evaluation, RAG & embedding retrieval (bge-m3), prompt architecture, agent benchmarking

## Experience

### AISpeech

Jun. 2026 – Aug. 2026

*Sponsored Capstone Thesis — LLM data flywheel for in-vehicle voice assistants*

Shanghai, China

- Built a synthetic-data pipeline that generated **17,200** verified training dialogues for the multimedia domain of an in-vehicle voice assistant, then **fine-tuned** the model on them — tool-call accuracy **72.8%→82.7%** on real user conversations, graded by an independent judge model (DeepSeek-V4-Pro), not the same model that scored the corpus.
- Built the user simulator behind it: **41** persona archetypes over a rule-based conversation planner plus LLM generation, grounded in the assistant's real tool index so every generated multi-turn dialogue exercised genuine tool calls rather than invented ones.
- Filtered every dialogue by reward-scoring it on a **397B** model, keeping only provably correct traces, and regenerated **200** dialogues for each of the **11** worst-performing functions — spurious no-tool calls fell **4,424→1,410**.

### ByteDance

May 2026 – Aug. 2026

*AI Solutions Intern; Stack: Python, Flask, React, TypeScript, Node.js, Playwright / CDP*

Shanghai, China

- Built an **LLM-agent evaluation platform** for two enterprise desktop agents: designed a suite of **200+** real user tasks, drove both Electron clients over the Chrome DevTools Protocol, normalized their incompatible telemetry into one trace schema, and scored them with an LLM-as-judge harness on output quality, token throughput and cost — one comparable read, and the team's basis for selection.
- Productized customer technical due diligence into a **citation-grounded RAG** research tool on internal AI tooling, where every conclusion resolves to its source; adopted by the regional client team as their working tool.
- Built an all-in-one **client-research agent** — deep company research, opportunity qualification, executive-engagement prep and meeting debriefs in a single pass — generating a one-page account playbook per client: a reasoned read of the company, hypothesized AI-adoption scenarios and a complete **FDE deployment proposal**; **2 h → 5 min** per account.
- Hand-coded all **11** agents' skills for a client-facing demo site — industry-specific and general-purpose — including a PPTX-to-HTML skill the team still uses daily, plus **13** case studies on a shared asset base of **240+** industry reference materials.

### Shenzhen Legion Technology Co., Ltd.

Apr. 2025 – Apr. 2026

*Data Engineering & AI Intern (part-time); Stack: Python, FastAPI, PostgreSQL, React/TS*

Remote, China

- Owned **MetaFlow** end to end (started Mar. 2026), a natural-language-to-SQL product (FastAPI / PostgreSQL / React): designed the metadata layer that routed business questions to the backend engine through a **verifiable intermediate layer** instead of raw SQL generation, and drove its iteration roadmap and role design — closing the loop from business users to data analysts to data engineers.
- Built its gold-case regression suite — **20** scenarios pinned across 5 pipeline stages as contract-as-test — plus a contracts endpoint running data-quality rules as executable checks, so a schema change was caught by re-running the suite, not by hand.
- Designed, earlier in the role, the prompt architecture of an open-source RAG data-governance copilot for dbt pipelines — question classification (catalog / SQL / data-QA), entity extraction, embedding-based semantic retrieval — and its evaluation framework of fixed test cases for retrieval accuracy and governance compliance, owning feature scope and release priorities, not the upstream codebase.

### PGF Technology Group

Mar. 2026 – Apr. 2026

*Consulting Engagement (4-person team), sole developer; Stack: Next.js, TypeScript, Postgres*

Rochester Hills, Michigan, USA

- Shipped a full-stack work-order tracking system as sole developer on a four-person consulting engagement, covering all **9** production stations of a ~50-person harness manufacturer — a complete solution the client took into internal use after handover.
- Diagnosed stale reads across serverless instances and migrated storage in production (in-memory → Blob → Postgres); then pinned down the scan, queue and alert flows with **6** Playwright e2e suites.

## Projects

### Cardiac Rehabilitation Wearable-Data Study

Apr. 2025 – Jan. 2026

*SURE Research Program, University of Michigan — Advisors: Profs. Mariel Lavieri & Jessica Golbus*

Ann Arbor, Michigan, USA

- Built a time-series feature pipeline over **400,000+** raw step records from **247** enrolled cardiac-rehab patients — rolling means, trend slopes, periodicity, plus imputation for gapped and high-variance series.
- Phenotyped recovery trajectories with K-means/PCA and detected plateaus with a Kalman filter: against the standard 36-session protocol, ~**68%** of the 191-patient analysis cohort was discharged off its true plateau.

### EMG-Driven Rehab Training Game & Muscle-Synergy Analysis

Oct. 2024 – Apr. 2026

*NeuRro Lab, University of Michigan — PI: Chandramouli Krishnan*

Ann Arbor, Michigan, USA

- Built the lab's EMG-driven rehabilitation training game **end to end** — Unity/C# game client and in-game UI, real-time signal pipeline, and the backend database — now a standard tool in ongoing clinical experiments.
- Engineered its real-time data path — **16-channel** EMG at **2 kHz** through band-pass, envelope extraction, EMA smoothing and MVC calibration — and ported its multi-objective gradient-descent optimizer from **PyTorch** to C# for on-device use across an **11-stage** protocol.
- Built the lab's muscle-synergy pipeline in MATLAB — NNMF-NNR, cross-subject alignment, statistics over **18** subjects; in clinical use.